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DEPARTMENT OF HEALTH PAPUA NEW GUINEA



CONTRACTORS ENVIRONMENTAL MANAGEMENT PLAN

PROJECT:	Construction and Completion of Community Health Post at Gaulim in Gazelle DISTRICT, EAST NEW BRITAIN PROVINCE		
EMPLOYER:	National Department of Health		
PROJECT:	Rural Primary Health Services Delivery Project		
COUNTRY:	Papua New Guinea		
DUE DATE:	14 th of July 2020		
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ISSUE 1:	For Client Approval		

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1. INTRODUCTION AND PURPOSE

1.1 INTRODUCTION

This Contractor's Environmental Management Plan (CEMP) details the implementation of measures in accordance with environmental commitments outlined in Environmental Management Plan (EMP). It will be is a contractual document which will be used by the Client as well as the Contractor (Haom Builders) to ensure the construction works complies with all the laws and regulations stated on Page 14 and 15 of the EMP. The personnel's involved in preparation of the CEMP are as follows;

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1.2 PURPOSE

The purpose of this CEMP is to outline how Haom Builders will avoid, minimize or mitigate the effects of construction work on the environment and the surrounding Gaulim Community area while undertaking the project. This CEMP will

- ✚ Highlight stakeholder requirement
- ✚ Ensure that the development is compliant with the current legislations
- ✚ Outline the EMP requirements in accordance with the laws and regulations
- ✚ Detail the mitigation committed to within the EMP and how it will be implemented on site
- ✚ Ensure that any adverse effects are minimized during construction
- ✚ Describe any site-specific method statement required

1.3 STRUCTURE

Since the project is defined as Level 1 under the Environmental Prescribed Activities Regulation (EPAR), this CEMP will be site specific as it will be focused on the environmental and geographical conditions of Gaulim area.

This document is structured as follows

- ✚ **Scope of works** - which will define the project and provide general information regarding the proposed development
- ✚ **Management responsibilities** - to identify the persons responsible for undertaking the work on behalf of the contractor
- ✚ **Legal Requirements** – states the laws, regulations and statutory requirements which safeguards the environment
- ✚ **Licenses and permits** – which identifies and includes consents, permission and commitments
- ✚ **Special Environmental or Cultural Issues** – identifies and locates protected areas and environment

- ✚ ***Machinery and Support Equipment Brought to site*** – states as declares the type of plants and equipment to use for the project which will more likely have an impact on the surrounding environment on site
- ✚ ***Details of Sites Used to Source raw materials*** – identifies and locates the sites which will be used for extraction of raw materials such as river gravel, water supply, timber etc
- ✚ ***Contractors Facilities and Workers Camps*** – details how Haom Builders will establish the site regarding workers accommodation, amenities, storage facilities
- ✚ ***Environmental Protection Work Procedures*** – declares the procedures designed to minimize the impact of construction activities to the environment
- ✚ ***Monitoring of Work*** – provides details on how each activity will be monitored on site
- ✚ ***Staff and Worker training*** – states the details of training and awareness on compliance of CEMP

2. SCOPE OF WORKS

The scope of works for the Gaulim Community Health Post will include the following construction activities;

- ✚ Mobilization
- ✚ Site Establishment
- ✚ Earthworks over site (Stripping of top soil & levelling)
- ✚ Construction of 1x Community Health Post Building
- ✚ Construction of 3x Staff Accommodations
- ✚ Construction of 1x Ablution Block, 1x Generator Shed and 1x Incinerator Shed
- ✚ Walkways and footpaths

2.1 MOBILIZATION

This will include mobilizing all the resources needed for the project to the project site. Mobilization for this project will begin with procurement of materials, plants, tools and equipment, recruitment and selection of skilled workers, organizing and transportation of all resources to project site.

This activity will be done away from the site and will not have an impact on the surrounding environment. However, all plants and equipment which will be committed to the project will be thoroughly checked and serviced to ensure that they are functioning well prior to being dispatched to the project site.

All the skilled workers will be selected via a proper recruitment process. Induction programme will also cover awareness on minimizing environmental damage on site. Mobilization stage will also consider the strategies to be implemented to safeguard the surrounding environment. All necessary resources for implementing this CEMP will be brought to site.

2.2 SITE ESTABLISHMENT

Site establishment is basically setting up the project site after mobilization and taking possession of site. This will involve erection of temporary facilities on site. Site facilities will include establishing site boundaries, hoarding and fencing, site office, storage shed, site accommodation, amenities, lay down area and temporary services (water supply, electricity supply, communication etc.).

These works will be done on the proposed site in Gaulim. Existing environment will be disturbed and altered. Site establishment will include erection of perimeter fencing and hoarding. Proper drainage systems for surface

run offs will be constructed as shown on the Proposed Site Plan (Sheet No: 03) prior to establishment of any structure on site.

All the site facilities will be contained within the site boundary. Negotiations with locals and project superintendent and safeguard specialist will be done for additional space for setting up of other site facilities if space on site is limited. Living quarters, ablution blocks, rubbish and waste disposal and containment areas will be located and constructed after all environmental protection measures are taken considered. Please refer to Appendix 01 (Contractors Site Establishment Plan) for the location of these facilities on site.

The following activities will be carried out during site establishment. Most of these activities will alter the existing environment. These activities will involve;

-  Slashing of existing vegetation
-  Surface excavation/top soil removal
-  Detail excavation
-  Trenching and drainage systems
-  Pit excavation
-  Backfilling and compaction
-  Concrete mixing and pouring
-  Felling of trees to obtain timber for construction of site facilities
-  Removal and disposal of excavated wastes
-  Access roads to site

The above activities may give rise to the following environmental issues on site;

-  Soil erosion
-  Loss of vegetation on top soil
-  Sedimentation of loose soil
-  Seepage of contaminants into water ways
-  Alteration of natural water ways
-  Noise and vibration
-  Dust generation

Mitigation of the above listed issues will be further discussed under Environmental Work Procedures. The above list can be expended further. However, they are introduced under this scope of work so that adequate mitigation procedures can be developed while taking each activity into account.

2.3 EARTHWORKS OVER SITE

Gaulim Community Health Post is located on a relatively flat area with the Community Health Post located at the highest point of site as and staff houses being located down the slope. The volume of earthworks will be be restricted the floor areas of each of the following buildings and structures; Main CHP Building, Ablution Block, Incinerator, Generator Shed, Tank Stands and Foot paths as they will be built on slab on ground. All three (03) Staff Houses will be built on galvanized stumps. This will slightly alter the level of top soil and vegetation within the building area.

Surface excavation will be limited to grubbing out the existing vegetation and improving of surface soil by means of compaction and consolidation of backfill materials. Highest points on site will be excavated to make

up levels. Excavated earth will be stock piled on site and reused to fill up other lower areas as well as filling to make up levels. Top/vegetable soil will be stockpiled to be reused for landscaping purposes.

Retaining walls, if required, will be constructed to contain soil from eroding from the faces of excavation. This will also protect the building area from buildup of sediments.

The generator shed, incinerator shed, and the ablution block will also require excavation and filling as they are also located on a gentle slope. Suitable excavated materials will be used to fill this area. Pits to the incinerator and septic to ablution block will be excavated.

Site access and egress points as well as traffic routes within the site will be excavated and filled with hardcore materials. Surface of all these routes will be compacted at an angle to promote efficient surface run off. Sides of all traffic routes within site will have spoon drains which connect to the main drain as shown on the site plan.

Destructive activities will obviously include;

-  Slashing of existing vegetation
-  Surface excavation
-  Bulk excavation
-  Trenching and drainage systems
-  Pit excavation
-  Extraction of hardcore fill material
-  Bulk filling and compaction
-  Concrete mixing and pouring (for retaining walls)
-  Removal and disposal of excavated materials
-  Access roads to working area

The following issues will most likely be needed to be addressed during earthworks;

-  Soil erosion
-  Loss of vegetation on top soil
-  Sedimentation of loose soil
-  Seepage of contaminants into water ways
-  Alteration of natural water ways
-  Noise and vibration
-  Dust generation
-  Damage to sites where raw materials are obtained

2.4 COMMUNITY HEALTH POST BUILDING

The proposed community health post building will be built on slab on ground which will increase the amount of excavation, filling and concreting. Therefore, major alteration will be done to the landscape and vegetation within the building perimeter.

The ramps in front of the entry porch will be concrete slab on concrete block walls. This element of the building will include concrete mixing and placing. Special considerations will be given to this area during concrete works.

The main structural framing members of the building will be galvanized steel framing. Factory coated galvanized steel frames will compose the walls, ceiling and roof.

External stair rails, entry porch rails and ramp rails will be metal framed utilizing primed hollow section steel tubes. Decking on entry porch will be timber.

Door heads and jambs will be dressed hardwood timber. Window heads, sills and mullions will also be hardwood timber.

Internal wall and ceiling lining will be fiber cement and villa board linings respectively. External claddings will be V-crimp sheets. Roof sheeting will involve corrugated roofing sheets.

Vinyl sheets, ceramic tiles and varnish will be used as finishes to floors on specific areas on the building. Epoxy paints will be used on external rails.

Most of the fixed furniture such as cupboards, benches and shelves will compose mainly of timber products. Most of these surfaces will need to be sanded and varnished.

Door and window hardware and accessories will be as per specification. Plumbing fittings will be Caroma fittings will stainless steel taps, spouts and accessories. These fittings will supply hot and cold water. Hot water will be supplied via overhead solar system. All drainage system will end up in a concrete septic tank

Destructive activities will obviously include;

-  Surface excavation
-  Bulk excavation
-  Pit and trench excavation to footings and septic tank
-  Bulk filling and compaction
-  Concrete mixing and pouring to footings and ramps
-  Removal and disposal of excavated materials
-  Cutting of CFC sheet wall and ceiling linings
-  Cutting of galvanized steel framing members
-  Cutting of timber products
-  Use of vinyl glue
-  Use of epoxy paints, varnish and other paints
-  Fabrication of steel framing members
-  Excessive use of electricity

The following issues will most likely be needed to be addressed during earthworks;

-  Soil erosion
-  Loss of vegetation on top soil
-  Sedimentation of loose soil
-  Seepage of contaminants (cement, paints, glue etc) into water ways
-  Noise from tools plants on site as well as cutting of metal
-  Dust generation during compaction of backfill materials, cutting and sanding timber products
-  Odour from paints and epoxy paints
-  Safety of workers and public

2.5 3x STAFF ACCOMODATION

The proposed staff houses will be built on galvanized stumps thus, limiting the amount of excavation, filling and concreting to pits of stumps. No major alteration will be done to the landscape and vegetation within the building perimeter.

The site will be terraced and retaining walls will be constructed to hold back materials that will be used to make up level and improve the surface condition of the soil.

The main floor framing members of the building will be hardwood timber. Wall, roof and ceiling frames will also utilize timber framing members.

External stair rails, entry porch rails and ramp rails will be metal framed utilizing primed hollow section steel tubes. Decking on entry porch will be timber.

Door heads and jambs will be dressed hardwood timber. Window heads, sills and mullions will also be hardwood timber.

Floor decking will be T&G flooring sanded and varnished to finish. Internal wall and ceiling lining will be plywood. External claddings will be V-crimp sheets. Roof sheeting will involve corrugated roofing sheets.

Most of the fixed furniture such as cupboards, benches and shelves will compose mainly of timber products. Most of these surfaces will need to be sanded and varnished.

Door and window hardware and accessories will be as per specification. Plumbing fittings will be Caroma fittings will stainless steel taps, spouts and accessories. These fittings will supply hot and cold water. Hot water will be supplied via overhead solar system.

Destructive activities will obviously include;

-  Surface excavation
-  Bulk excavation
-  Trenching and drainage systems
-  Pit excavation
-  Bulk filling and compaction
-  Concrete mixing and pouring to footings and retaining walls
-  Removal and disposal of excavated materials
-  Cutting of timber products
-  Use of epoxy paints, varnish and other paints
-  Excessive use of electricity

The following issues will most likely be needed to be addressed during earthworks;

-  Soil erosion
-  Loss of vegetation on top soil
-  Sedimentation of loose soil
-  Seepage of contaminants (cement, paints, glue etc) into water ways
-  Noise from tools plants on site as well as cutting of metal
-  Dust generation during compaction of backfill materials, cutting and sanding timber products

- ✚ Odour from paints and epoxy paint
- ✚ Safety of workers and public

2.6 CONSTRUCTION OF ABLUTION BLOCK, INCINERATOR SHED & GENERATOR SHED

All these three (03) supports facilities will be of concrete construction. These buildings will incorporate reinforced concrete slab on ground. Ablution block will have full height concrete block walls. Generator shed, and the incinerator shed will have block walls terminating at specified heights. Primed hollow section steel tubes will be anchored onto the block walls to make up the remaining height of walls. Roof frames will be galvanized steel with corrugated roofing sheets.

Destructive activities will obviously include;

- ✚ Surface excavation
- ✚ Bulk excavation
- ✚ Trenching and drainage systems
- ✚ Pit excavation
- ✚ Bulk filling and compaction
- ✚ Concrete mixing and pouring to slabs and core fill to block walls
- ✚ Removal and disposal of excavated materials
- ✚ Cutting of concrete products
- ✚ Use of epoxy paints, varnish and other paints
- ✚ Excessive use of electricity

The following issues will most likely be needed to be addressed during earthworks;

- ✚ Soil erosion
- ✚ Loss of vegetation on top soil
- ✚ Sedimentation of loose soil
- ✚ Seepage of contaminants (cement, paints, glue etc) into water ways
- ✚ Noise from tools plants on site as well as cutting of metal and concrete blocks
- ✚ Dust generation during compaction of backfill materials, cutting concrete products
- ✚ Odour from paints and epoxy paint
- ✚ Safety of workers and public

2.7 WALKWAYS AND FOOTPATH

Footpaths/walkways will compose most of the external works. This will mainly involve groundworks and concrete works. These two works will also have an impact on the natural environment on site and surrounding as they will be generating harmful wastes. Weight of concrete on slab may cause minor landslip on site as the site is prone to water seepage. Water from higher points uphill normally seeps out at the community health post area.

Destructive activities will obviously include;

- ✚ Surface excavation
- ✚ Bulk excavation
- ✚ Trenching and drainage systems

- ✚ Pit excavation
- ✚ Bulk filling and compaction
- ✚ Concrete mixing and pouring to slabs and core fill to block walls
- ✚ Removal and disposal of excavated materials
- ✚ Cutting of concrete
- ✚ Excessive fume emission from concreting plants

The following issues will most likely be needed to be addressed during earthworks;

- ✚ Soil erosion
- ✚ Loss of vegetation on top soil
- ✚ Sedimentation of loose soil
- ✚ Seepage of contaminants (wet cement) into water ways
- ✚ Noise from tools plants on site
- ✚ Dust generation during compaction of backfill materials, cutting concrete products
- ✚ Odour from paints and epoxy paint
- ✚ Excessive fume emission
- ✚ Safety of workers and public

3. MANAGEMENT RESPONSIBILITIES

The persons identified in the following tables will be assigned and attached to form the site management team for the Gaulim Community Health Post Project. They will be directly responsible for the CEMP activities. The Site Manager is the most superior person on site followed by the Site Supervisor. He will ensure that all the works and activities on site conform to the CEMP. The organizational chart on the following page illustrates the management structure, the proposed channel of communication and authority on management of the CEMP on site.

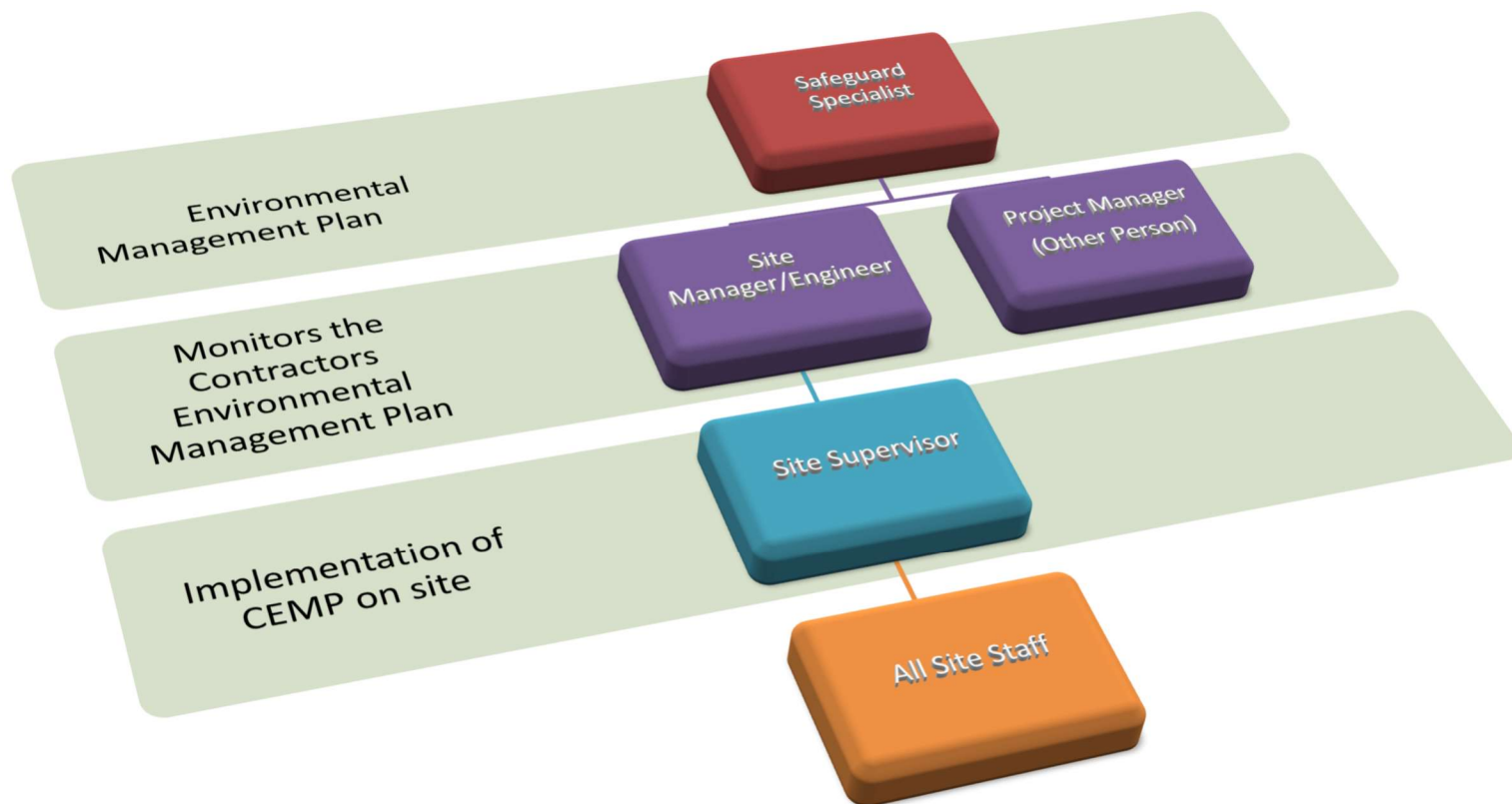


Figure 1. Organizational Structure for the onsite management of CEMP

These persons may be replaced by other persons after submission of this document. The officer in charge will be made aware of such changes at the earliest juncture.

3.1 ENVIRONMENTAL/SITE MANAGER

Name	Position	Availability on site	Office	Mobile	Email
Mr. Hamo Anep	Environmental / Site Manager	Full Time	472 0601	72322660	haombuilders@gmail.com

3.2 BACK UP PERSON

Name	Position	Availability on site	Office	Mobile	Email
Mr. Babayu Kerry	Site Supervisor	Full Time	472 0601	74211179	haombuilders@gmail.com

3.3 OTHER PERSONS

Name	Position	Availability on site	Office	Mobile	Email
Mr. Cherry Rauna	Construction Manager	Part Time	472 0601	79393365	haombuilders@gmail.com

4. LEGAL REQUIREMENTS

We, Haom Builders, understand that it is our responsibility to ensure that all the works carried out on site is within the jurisdiction of various environmental laws, regulations and standards mentioned in this CEMP. We understand that we will always seek clarification from these documents should any conflict arise during work. We acknowledge the following legal documents to form part of the legal requirements along with this CEMP;

- 📄 ADB Safeguard Policy Statement
- 📄 Environmental Act 2000
- 📄 Environmental Prescribed Activities Regulations
- 📄 Land Group Incorporation Act 1974
- 📄 Land Registration Act 1981

5. SPECIAL ENVIRONMENT OR CULTURAL ISSUES

Local knowledge will be sought if we come across anything of cultural significance is located on site during work. There is a river close to the site. However, measures will be taken to prevent spills of pollutants into the river. Plants and vegetation immediately outside the perimeter of the site will be protected as much as possible. All surface run offs will be diverted into soak up pits on site. Contaminated spills and liquid wastes will also be disposed in separate containment/soak up pits.

River gravel for filling and surface improvement on site will be sourced from a river near project site. Backhoe will be used to dig out gravel, stockpile and load onto dump truck to be transported to site

Since the materials will be quarried from river and, the materials harvested will be spread along a wide area to prevent pooling and diversion of river. We will make sure that minimum damage is made to the quarry and

the surrounding vegetation. Likelihood of landslips will be minimized. Stockpiles will be done at the base of the excavated area.

Existing buildings will be protected from being damaged. Barricades will be put in place to identify the boundary of the building. Compaction of the main site areas will be addressed thoroughly to prevent disruption to existing buildings foundation.

6. MACHINERY AND SUPPORT EQUIPMENT BROUGHT TO SITE

The following table shows the details of machineries and equipment that we intend to commit to the project. Please refer to Appendix No: 02 for some of the photos of these equipment's.

PLANT NAME	MAKE & TYPE	REGO	AGE	CONDITION	CURRENT LOCATION
1x 3m ³ Ton Dump Truck	Isuzu	LBF 211	5 yrs. old	Recently serviced and running – Very Good Condition	Lae, East New Britain Province
1x Land Cruiser (Utility)	Toyota	LBF 590	5 yrs. old	Recently serviced and running – Very Good Condition	Lae, East New Britain Province
2x 5KVA Generator	Power trade		Brand New	Very Good Condition	Lae, East New Britain Province
2x 300L concrete mixers	Truemax		Brand New	Very Good Condition	Lae, East New Britain Province
1xConcrete vibrator	Truemax		Brand New	Very Good Condition	Lae, East New Britain Province
30x Scaffold set			Brand New	Very Good Condition	Lae, East New Britain Province
2x Portable welding machine – 175 inverter	Weld master		1 year old	Very Good Condition	Lae, East New Britain Province
1xAutomatic Dumpy Level	Sokkia		1 year old	Very Good Condition	Lae, East New Britain Province
1xTamping Rammer	Truemax		6 months old	Very Good Condition	Lae, East New Britain Province
1xVibratory Plate Compactor	Truemax		6 months old	Very Good Condition	Lae, East New Britain Province

7. DETAILS OF SITES USED TO SOURCE RAW MATERIALS

7.1 GRAVEL

We have identified sites, along the banks of the Warangoi river, for extraction of gravel. This gravel will be used for improving surface condition of soil. The sites are less than a kilometre from the construction area. Access to the gravel pits are good condition. The locals normally get gravel for their building projects in the area.

7.2 WATER

Water for construction work and drinking will be piped separately from nearby water tap up the mountain. Storage tanks will be kept for rainwater in case of water shortage. Additional water for construction works can be pumped from the river if necessary or during dry season.

7.3 BUILDERS MIX

Builders mix will be sourced from local suppliers and transported to site. Suppliers will utilize their own plants and equipment to extract and deliver materials to site.

8. CONTRACTORS FACILITIES AND WORKERS CAMPS

The Table below contains the general scope of site facilities. Please note that the floor areas of the facilities may decrease due to space limitations or increase due to functional requirement on site. Appendix No: 01 shows the location of these facilities on site.

FACILITY	AREA (m ²)	CONSTRUCTION			STORM WATER DRAINS TO:	EFFLUENT DRAINS TO:	ELECTRICITY SUPPLY SOURCE	COLD WATER SOURCE
		FLOOR	WALLS	ROOF				
Site Office - Project manager's office - Site Engineer's Office - Meeting Room - Toilet - Tea room	72m ² (12m long x 4m wide)	Timber framed on low post, 18mm ply boards	Timber framed with V-crimp to external walls & 4mm plywood to internal walls	Timber framed with corrugated roofing sheets	Sediment basin (natural drainage system)	Fiber glass septic tank for 5 persons	5kva Genset	9000L TUFFA Tank & existing supply
Site Manager's Accommodation & Site Engineers Accommodation	48m ² (12m long x 4m wide)	Timber framed on low post, 18mm ply boards	Timber framed on low post, 18mm ply boards	Timber framed with corrugated roofing sheets	Timber framed with corrugated roofing sheets	Fiber glass septic tank for 15 persons	5kva Genset	9000L TUFFA Tank & existing supply
Workers Accommodation	90m ² (15m long x 6m wide)	Compacted river gravel base/floor	Timber framed with V-crimp to external walls & 4mm plywood to internal walls	Timber framed roof with corrugated roofing sheets	Sediment basin (natural drainage system)			
Workers Kitchen & messing facilities	36m ² (9m long x 4m wide)	Compacted river gravel base/floor	Timber framed with V-crimp to external walls & 4mm plywood to internal walls	Timber framed roof with corrugated roofing sheets	Sediment basin (natural drainage system)			
Shower rooms and laundry area	18m ² (6m long x 6m wide)	Reinforced Concrete floor	Timber framed with V-crimp to external walls & 4mm plywood to internal walls	Timber framed roof with corrugated roofing sheets	Sediment basin (natural drainage system)			
Material & Small plants & Equipment Storage Shed	100m ² (10m square)	Compacted river gravel base/floor	Timber framed with V-crimp to external walls & 4mm plywood to internal walls	Timber framed roof with corrugated roofing sheets	Sediment basin (natural drainage system)	NA	NA	NA
Vehicle and plants lay down area	100m ²	Compacted river gravel base/floor	NA	NA	Sediment basin (natural drainage system)	NA	NA	NA

9. ENVIRONMENTAL PROTECTION WORK PROCEDURES

9.1 SITE PREPARATION

-  Top soil will be altered by means of mechanical excavation
-  Exposed surface of soil may be subjected to erosion and sedimentation at lowest points.

Objective	To appropriately plan and prepare site so that damage to environment on site is minimum.		
Actions	Requirements	Responsibility	Timing
	No marine or terrestrial vegetation will be disturbed without prior approval	Site Manager, Site Supervisor	Throughout construction works
Performance Indicators	No injury / harm to local flora and fauna	Site Manager, Site Supervisor	Throughout construction works
Monitoring	Visual monitoring	Site Manager, Site Supervisor	Throughout construction works
Reporting	Reporting to Site Manager and SS	Site Manager, Site Supervisor	Throughout construction works
	Any reporting to regulators	Site Manager, Site Supervisor	
Corrective Actions	Corrective actions and implementation time frame	Site Manager, Site Supervisor	Throughout construction works

9.2 EXCAVATION OF CONSTRUCTION SITES

-  Potential sources of sediments
-  Risks of erosion
-  Stockpiles
-  Stormwater discharge points
-  Existing field and kerb inlets and other stormwater infrastructure
-  Site access and egress locations
-  Nearby sensitive areas

Objective	To manage construction activities which exhibit the potential to cause erosion and release of sediment		
Actions	Requirements	Responsibility	Timing
	Minimize erosion and water ponding Minimize injuries to workers	Site Manager, Site Supervisor	Prior to construction commencing
Performance Indicators	No mud/dirt visible on construction site		Throughout construction works
Monitoring	Daily visual inspections of works site and all erosion and sediment controls	Site Manager	Throughout construction works

Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Morobe Provincial Health & Works	SS	
Corrective Actions	Corrective actions and implementation time frame	Site Manager, Site Supervisor	Throughout construction works

9.3 REMOVAL AND DISPOSAL OF EXCAVATED SOIL

Include:

-  All excavated materials will be reused to fill up lower areas a retaining wall
-  Top soil will be stored and used for landscaping works

Objective	To appropriately manage excavated materials over site and minimize disposal off site.		
Actions	Requirements	Responsibility	Timing
	Suitable location will be identified for storage of excavated materials. Materials will be separated and reused appropriately	Site Manager	Prior to works commencing
Performance Indicators	Minimize unnecessary stockpile of materials and erosion of materials	Site Manager	Throughout remediation works
Monitoring	Daily inspection of storage area	Site Manager	Throughout remediation works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Morobe Provincial Health & Works	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

9.4 EROSION AND SEDIMENTATION

Include:

-  Potential sources of sediment/risks of erosion
-  Stockpiles

- ✚ Stormwater discharge points in open drains and soak up pits
- ✚ Existing field and kerb inlets and other stormwater infrastructure
- ✚ Site access and egress locations
- ✚ Nearby sensitive areas
- ✚ Diversions for surface run offs

Objective	To manage construction activities with the potential to cause erosion and the release of sediment		
Actions	Requirements	Responsibility	Timing
	Construct sumps or absorption pits along certain points of drains to keep eroded materials	Site Manager	Prior to construction commencing
Performance Indicators	Minimize erosion over site	Site Manager	Throughout construction works
Monitoring	Daily visual inspections of works site and all erosion and sediment controls	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Morobe Provincial Health & Works	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

9.5 NOISE AND VIBRATION

Include:

- ✚ Noise generating activities such as steel fabrication will be done off site.
- ✚ Maintenance and operation of machinery / equipment / vehicles
- ✚ Work hours
- ✚ Specify any activities to be undertaken at night and why (if any)

Objective	To appropriately manage noise during construction activities to minimize impact to workers, neighbors and community members.		
Actions	Requirements	Responsibility	Timing
	all significant noise generating activities will be undertaken between Mon-Sat 6.30am to 6.30pm	Site Manager	Throughout construction works
Performance Indicators	no complaints regarding noise from workers / neighbors / community members	Site Manager	Throughout construction works
Monitoring	noise monitoring program or complaint-based monitoring	Site Manager	Throughout construction works

Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Morobe Provincial Health & Works	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

9.6 DUST GENERATION

- ✚ Include the following:
- ✚ Dust suppression
- ✚ Stockpiles
- ✚ Truck loads
- ✚ Maintenance and operation of machinery / equipment / vehicles
- ✚ Other methods to prevent / minimise dust

Objective	To manage construction activities with the potential to impact on air quality		
Actions	Requirements	Responsibility	Timing
	water carts will be used for dust suppression	Site Manager	Throughout construction works
Performance Indicators	no complaints regarding dust from workers / neighbors / community members	Site Manager	Throughout construction works
Monitoring	air quality monitoring program or complaint-based monitoring	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Morobe Provincial Health & Works	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

9.7 PUBLIC ACCESS TO SITE

Include the following:

- ✚ Installation of proper signage barricades
- ✚ Logbooks and entry pass
- ✚ Security at all entry and exit point

Objective	To manage construction activities with the potential to impact on air quality		
Actions	Requirements	Responsibility	Timing

	Site will be accessed via guard house	Site Manager	Throughout construction works
Performance Indicators	No trespasser on site.	Site Manager	Throughout construction works
Monitoring	Security checks at entry points	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Morobe Provincial Health	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

9.8 RISK TO PUBLIC AND WORKER HEALTH AND SAFETY (OHS)

Include the following:

-  Provide PPEs
-  PPE checks
-  Truck loads
-  Warning signs
-  Barricades
-  Maintenance and operation of machinery / equipment / vehicles
-  Other methods to prevent / minimise dust

Objective	To manage construction activities with the potential to impact on air quality		
Actions	Requirements	Responsibility	Timing
	All workers will be to have all necessary PPEs	Site Manager	Throughout construction works
Performance Indicators	No injuries or accidents on site	Site Manager	Throughout construction works
Monitoring	PPE check every morning during toolbox meeting	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Any reporting to Department of Environment and Heritage Protection	SS	
Corrective Actions	Corrective actions and implementation time frame	Site Manager	Throughout construction works

9.9 USAGE AND STORAGE OF HAZARDOUS MATERIALS

-  Storage
-  MSDS's

- ✚ Refuelling
- ✚ Spill kits and emergency spill response
- ✚ Spill kit training
- ✚ Wrapping or covering of powdered materials (cement, grouts etc)

Objective	To manage construction activities with the potential to cause contaminated soil and harm workers		
Actions	Requirements	Responsibility	Timing
	All hazardous will be stored in secured, roofed areas. Use of canvas or Visqueen, sawdust or gravel bedding for fuel storage/refueling areas to be replaced if spillage is encountered.	Site Manager	Throughout construction works
Performance Indicators	No spills or injuries to workers	Site Manager	Throughout construction works
Monitoring	chemical storage areas and spill kits will be inspected on a routine basis	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Any reporting to Department of Environment and Heritage Protection	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

9.10 DISPOSAL OF WASTE MATERIAL (SOLID AND LIQUID)

Include:

- ✚ Waste burial pits organic materials
- ✚ Disposal bins for inorganic materials

Objective	To appropriately manage waste during construction activities.		
Actions	Requirements	Responsibility	Timing
	Appropriate waste receptacles will be provided on site	Site Manager	Prior to works commencing
Performance Indicators	No waste deliberately or unintentionally released	Site Manager	Throughout remediation works

Monitoring	Reporting to Site Manager and SS	All staff	Throughout remediation works
Reporting	Any reporting to Department of Environment and Heritage Protection	SS	Throughout construction works
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

9.11 ARCHAEOLOGICAL DISCOVERIES

Include:

- ✚ Any known areas of Cultural Heritage Significance
- ✚ Steps to be undertaken if cultural heritage artefact is uncovered (“cultural heritage duty of care”)

Objective	To appropriately manage construction activities with the potential to disturb / uncover cultural heritage sites / Artifacts.		
Actions	Requirements	Responsibility	Timing
	All workers will be informed of anything of cultural significance on site	Site Manager	Throughout construction works
Performance Indicators	No cultural heritage artifacts uncovered or damaged	Site Manager	Throughout construction works
Monitoring	Visual monitoring	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Any reporting to regulators	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

9.12 REHABILITATION OF CONSTRUCTION SITES AND DEMOBILIZING OF CONTRACTOR FACILITIES

Include:

- ✚ Schedule maintenance of site
- ✚ Housekeeping over site
- ✚ Removal of contaminated soil
- ✚ Removal of solid wastes

Objective	To appropriately manage construction site so that minimum damage is done to environment close to the proximity of site.		
Actions	Requirements	Responsibility	Timing

	Maintenance of site facilities and housekeeping during construction period.	Site Manager	Throughout construction works
Performance Indicators	Well organized and kept site with no rubbish or contaminants on soil	Site Manager	Throughout construction works
Monitoring	Visual monitoring	Site Manager	Throughout construction works
Reporting	Reporting to site manager and SS	All staff	Throughout construction works
	Any reporting to regulators	SS	
Corrective Actions	Corrective actions and implementation time frame	Site Manager	Throughout construction works

10. MONITORING OF WORK

Monitoring of CEMP will be done progressively daily. Two (02) documents will be used to with the CEMP to monitor environmental issues on site. The CEMP will be used together with our Trade Specific Method Statement and Project Site Diary. Samples of each of these documents are attached in Appendix No. 03 and No: 04.

The method statement will outline how each activity in each of the working trades will be carried out while considering potential risks to both the environment and the progress of work itself. The method statement will be issued to the Safeguard Specialist (SS) before commencement of work.

For example, if we are to commence concrete works after the completion of groundwork's, a fully furnished method statement will be issued a week prior to the commencement of work on a trade. The method statement will state how all the activities in this trade will be carried out in sequence. Considerations will be given to the nature of impact each of the activities will have on the environment. This potential impact will be identified as a risk under Part 6 of the Trade Specific Method Statement.

The site engineer and the supervisor will ensure that the risk mitigating factors on the method statement are executed on site. The Project Manager and the Site Engineer will ensure that the SS is well informed on any changes in the method statement.

Site Log Book will also be used to record all activities on site each day. It will also have provisions where all activities relating to environmental protection impact mitigation will be recorded. The site engineer will be responsible for keeping record in the Project Site Logbook.

Monthly reports on CEMP activities will summarise all the records on the daily site dairy. It will also status of all activities in the method statement and CEMP. The Project Manager and the Site Engineer will be responsible for compilation of the report.

11. STAFF AND WORKER TRAINING

All the site personnel recruited for the Community Health Post Projects will go through induction. They will be made aware of the CEMP requirements prior to mobilisation to site. Our standard procedure for

every project site is organising a Tool Box Meeting every morning prior to work. From this tool box meeting, awareness will be made concerning various areas of concern in the CEMP.

When commencing work on new trade area, all site personnel will congregate to an assembly point to be informed about the risks and the impacts that trade area will have on the environment. They will be informed on waste management and rubbish disposal strategies that are in place. The project manager and the site engineer will work in collaboration with the Safeguard Specialist. Awareness may be in form of verbal explanation, pictorial illustrations, and video and power point presentations.

The CEMP reporting structure will also be made clear so that workers will know whom they will report to in terms of CEMP requirement. Monthly or weekly reports as agreed will be submitted to the SS and SO for review of the CEMP if necessary.

12. GRIEVANCE REDRESS MECHANISM

All grievances concerning the environment will be addressed by the Site Manager. However, we will also involve our managing director who is also responsible for public relations of the company in East New Britain Province. We will utilize the Grievance Intake Form attached in Appendix 1 in the EMP. Permission will be sought from safeguard specialist or concerned authorities.

Other complaints regarding the project itself will be handled by the Site Manager and the Public Relations Officer (PRO). Our PRO in East New Britain Province is the Managing Director himself. He has overseen the addressing grievances in some of our past projects in East New Britain Province.

Complaint forms will also be provided at the site office. The project complaint form will be a different form the Grievance Intake Form. Complainants will write down their complaints on the forms. All complaints will be referred to the Site Manager and the PRO who will then assess and give feedback the way they see appropriate.

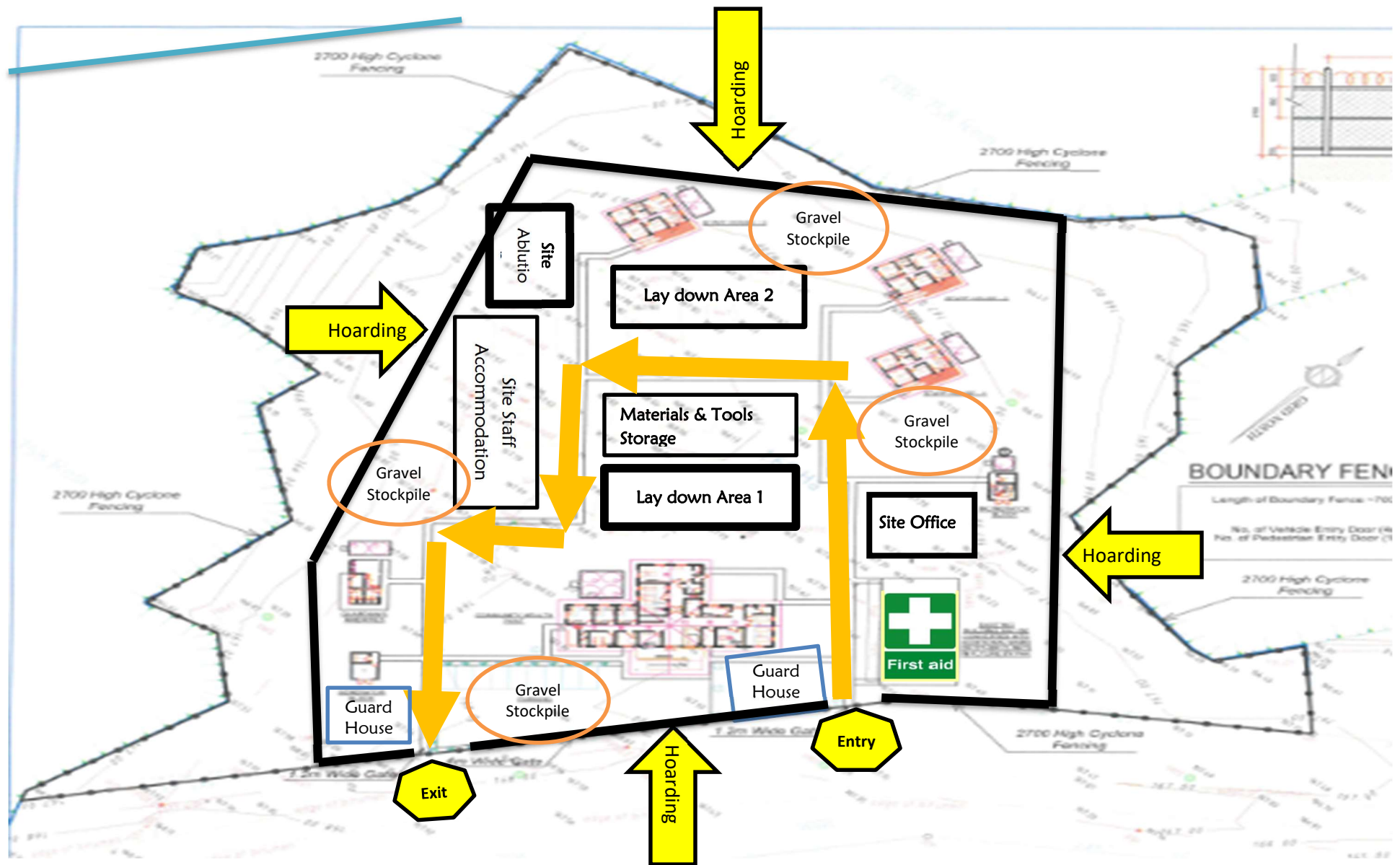
The proposed procedure in addressing complaints/grievances concerning the environment is illustrated by the process chart on the following page.



Figure 2. Steps to be taken to address grievances on site

13. APPENDICES

13.1 Appendix No: 01. CONTRACTORS SITE ESTABLISHMENT PLAN



13.2 APPENDIX NO: 02 PHOTOS OF MACHINERY & SUPPORT EQUIPMENT THAT WILL BE BROUGHT TO SITE



3m³ Dump truck



Toyota Land Cruiser



Isuzu NPS 17 Tonne Flat Tray



One of the 5KVA generators to be used for the Watut CHP



300L cement mixers bought and ready to be brought to site.



Newly bought tamping rammer & vibratory plate compactor



MITRE radial cross cut saw for cutting timber

13.3 APPENDIX NO: 03 SAMPLE OF TRADE SPECIFIC METHOD STATEMENT No: ONE (01) – COMMUNITY HEALTH POST

PART 1: PROJECT DETAILS

TRADE SPECIFIC METHOD STATEMENT NO:	HB001	DATE	
SITE ADDRESS	Community Health Post		
PROJECT SITE	Gaulim	PRINCIPLE	National Department of Health
PROJECT NAME	Rural Primary Health Services Delivery Project		
CONTRACT NO: East New Britain - 05/2021		CLIENT	RURAL PRIMARY HEALTH SERVICES DELIVERY PROJECT OF NATIONAL DEPARTMENT OF HEALTH FOR EASTN NEW BRITAIN PROVINCIAL HEALTH AUTHORITY
PROJECT MANAGERS	National Department of Health Commercial Support Services		

PART 2: CONTRACTOR DETAILS

CONTRACTOR	Haom Builders	ADDRESS	P.O. Box 797, LAE East New Britain Province
PROJECT MANAGER	Mr. HAMO ANEP	QUALIFICATION	BBLD.UoT
SITE MANAGER	Mr. Greg Collin	QUALIFICATION	BBLD.UoT
SITE SUPERVISOR	Mr. Babayu Kerry	QUALIFICATION	Tradesman Certificate

PART 3: SCOPE OF WORKS & WORK BREAK DOWN

BUILDING ELEMENT		Substructure		BUILDING LEVEL		Ground Floor Level	
WORKS INVOLVED		Groundwork's		DURATION			
ACTIVITIES INVOLVED		MAIN MATERIAL(S)		PLANTS/EQUIPMENTS		SKILLS REQUIRED	
		DESCRIPTION	QTY	DESCRIPTION	QTY	DESCRIPTION	QTY
1	Detailed Excavation Below Ground Floor Level for: Stripped Footings Pad Footings	Excavated material	98.4m ³	Automatic Dumpy Level & staff	2	Driver	1
				Building tape	5	Carpenters	24
				String lines 100mm	10	TA Carpenters	12
				Chalk line	10		
				Spades	20		
				Crew bar	5		
				Wheel barrows	10		
				Hydraulic rammer	2		
2	Maintaining Faces		109 m ²	Cutting blades	10		

3	Surface Preparation Trimming of sides, leveling of surfaces	Battered, trimmed, graded and compacted surfaces	212 m ²				
4	Hardcore filling & Compaction Placing of approved hardcore fill material & compact to required dry density.	Diorite, Road base or equally approved backfill material	188.6m ³				
5	Soil Treatment	Subcontractor to provide materials	574 m ²				
6	Moisture Membrane	Polythene membrane	459 m ²				

SEQUENCE OF WORK

Excavation of foundations below ground floor level will commence once profiles area confirmed by the engineer and the Superintendent.

Profiling and setting out for excavation will be done once all site establishment works are completed. Surface to be excavated should be marked with chalk line.

All excavation will be done manually as the sizes of footings do not allow mechanical means of excavation. Maintenance of faces after mechanical means will be additional work.

All excavated material will be stock piled 1m or 2m away from the trenches and removed by the backhoe.

Sides and faces of excavation will be trimmed and leveled. All loose soil will be removed from the surface.

Back fill materials will be transported to specific work areas using wheel barrows and placed.

Compaction will be done by hydraulic rammer to specified dry density. Compaction will be done after every 150mm.

Heights and levels will be maintained by taking levels using dumpy level and height pegs.

Soil treatment will be carried out by a Specialist Subcontractor.

Moisture membranes will be laid and lapped as specified. All ends will be securely fastened to the soil or other supports.

All loose soil or rubbish on the surface of moisture membranes will be removed before insertion of reinforcement.

PART 4: ADDITIONAL SUPPORT/STRUTTING

Temporary supports and propping/strutting will be constructed to prevent erosion of sides of excavation. Barricades will be erected on weak soil around the perimeter of excavation. Opposite is an example of how we intent to construct our support if we see necessary.

PART 5: METHOD OF ACCESS AND EGRESS

ACCESS TO GROUND FLOOR LEVEL	Barricades will be erected around excavated area. Entry points of small plants and employees will be established. Traffic routes will be established away from profiles and excavation. No employee will enter the excavated area without authorization from Engineer or Supervisor. Heavy machines such as trucks will not be allowed close to edge footing excavation. All materials into the foundations will be transported via wheel barrows to prevent loose edges of excavation from being damaged.
WORKING SPACES	Working spaces will not be allowed in trench excavation for footings unless there is weak soil to support. 400mm will be the minimum allowable working space.

PART 6: RISK IDENTIFICATION

POTENTIAL RISKS	POTENTIAL IMPACT ON WORK	COUNTER MEASURES
BAD WEATHER The site experiences unpredictable weather patterns with prolonged rainy season all throughout the year.	Delay in delivery of materials on site	Seek information on daily weather forecasts from Provincial Weather Office or reliable sources so that it should be considered in planning and execution of work on site.
	Water logging over site and in bottom of excavated surface.	All excavated surface of the trenches will be covered with canvas. If waterlogged, water will be pumped out using buckets or fuel generated pumps.
	Surfaces of excavation may erode & sediments may build up at the lowest point of site in front of the staff residences	An absorption pit will be excavated at the lowest point of the site to receive water and sediments from all drains.
	Work will be stopped on rainy days which may affect work progress and schedule	No electrical tools will be needed during this work. Canvas canopy will be constructed above working area, so work continues. Proper drainage will be done to remove excessive water.
ACCIDENTS & INJURIES ON SITE An employee or a member of the public would be injured.	Work would be stopped or delayed.	Implement Haom Builders Safety Policy on site and monitor the policy on site.
		Tool box meetings will be held every morning for safety checks and briefing
		Temporary fence and hording will be erected. Entrance and exit gates will be guarded.
		Establish traffic management on site and designate routes to work areas of employees, excavation plants and delivery vehicles.
		Erect barricades and signs indicating potential danger on site.
		No vehicle or other plants routes will be located close to the edge of excavation.

		Temporary supports and strutting will be erected on weaker soil on the surface around the perimeter of excavation.
		Only qualified operators will be allowed to operate hydraulic rammers. No worker will be allowed to stand close to the machine.

PART 7: HARZARD IDENTIFICATION

HEALTH HARZARDS DURING EXCAVATION OF FOUNDATIONS ON GROUND FLOOR LEVEL		MITIGATION OF HARZARDS
1	Noise pollution from excavators and vehicles when in use.	Hoardings will be erected as a noise barrier.
		Engines of all plants should be switched off when at rest.
		Ear muffs will be issued to every employee on site
2	Risk of falling into excavation area	Hoarding will be erected around the perimeter of excavation. Temporary supports will be constructed to hold loose soil on excavated sides in place.
		No employee or unauthorized person will be allowed close to excavation or into where excavation is done.
3	Dust pollution due to moving machines and work	Dust masks, safety glasses and other PPEs will be issued to each employee on site
		Hoardings will be erected as a dust barrier.
4	Traffic on site	Implement onsite traffic management plan

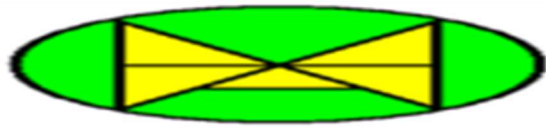
PART 8: SAFETY MANAGEMENT ON SITE

SAFETY POLICY	EMERGENCY PROCEDURES	FIRST AID FACILITIES	
Haom Builders Safety policies include an Onsite Safety Management Plan which sets out how safety measure will be implemented, monitored and enforced on site. Please refer the Safety Policies and Management Plan	Please refer to Haom Builders Safety Policies and Management Plan	Safety Officer	Moses Yambu
		First Aid Office Location Indication	
		Nearest Hospital	Vunapope Hospital

PART 9: PERSONNEL PROTECTION EQUIPMENT ON SITE

PERSONNEL PROTECTION EQUIPMENT TO BE USED FOR EXCAVATION WORKS	
 EYE PROTECTION MUST BE WORN IN THIS AREA  HEARING PROTECTION MUST BE WORN IN THIS AREA  DUST MASK MUST BE WORN IN THIS AREA  HEAD PROTECTION MUST BE WORN IN THIS AREA  FOOT PROTECTION MUST BE WORN IN THIS AREA  HIGH VISIBILITY CLOTHING OR VEST MUST BE WORN IN THIS AREA	
EMERGENCY PROCEDURES Please refer to our Health and Safety Policies	FIRE – If minor, immediately sound alarm and alert all those present within the premises and tackle same with onsite fire extinguishers. Call fire service if not brought under immediate control. If major, evacuate building and call Provincial Fire Service. Report ALL accidents to site safety officer and Project Manager. Notify Safety officer to investigate. ACCIDENT – Report immediately to Site Engineer and Project Manager. Action to be taken as per Haom Builders Health and Safety Policy.

13.4 APPENDIX NO: 04 PROJECT SITE LOGBOOK



**P.O.Box 797, LAE 411
Morobe Province
Papua New Guinea**

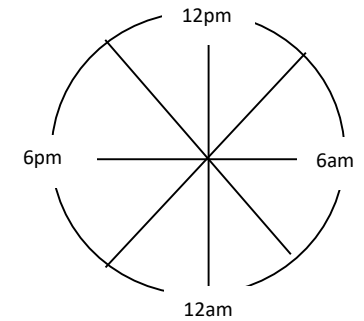
HAOM BUILDERS

**Phone: 472 0601
Fax: 472 0625
Mobile: 7160 0400 (Digicel)
E-Mail: haombuilders@gmail.com**

SITE DIARY

PROJECT:	Gaulim Community Health					
CLIENT:	EAST NEW BRITAIN PROVINCIAL HEALTH AUTHORITY					
CONTRACT TYPE:	MAJOR WORKS					
MAIN-CONTRACTOR:	HAOM BUILDERS					
CONTRACT VALUE (PGK):						
DATE:						
SITE ENGINEER:	HAMO ANEP					
MANPOWER				MACHINERY		
DESCRIPTION	MAN/ DAY	HRS	COST CODE	DESCRIPTION	MACHINE PER DAY	HOUR
Site Manager				Tipper Truck		
Supervisor				Backhoe		
Carpentry Foreman				Front End Loader		
Leading Hand Carpenters				Bulldozer		
Carpenter				Excavator		
Plumbing Foreman				Drum Roller		
Plumber				Generator		
Joiner Foreman				Plate Compactor		
Joiner				Vibrator		
Aluminum Fixer				Tampering rammer		
Boilermaker				Water Pump		
Welder						
Rigger				WEATHER CHART		
Steel Fixer				FAIR	1	
Gyprock Plasterer				CLOUDY	2	

Concrete Finisher				MODERATE RAIN	3
Tiler				HEAVY RAIN	4
Painter				STORMY	5
Electrical Sub-Contractor					
Storeman					
Site Security					
Laborers					



SUMMARY OF WORKS UNDERTAKEN:

AREAS OF CONCERN:

Checked by Project Manager: _____ Signature: _____